Original Article

Optimisation of HDPE-PU Thermal Insulation: Effect of Processing Parameters on Insulation Properties

Joshua A. Gidigbi[1,*] | Sunday A. Osemeahon[1] | Abdulrasaq B. Abubakar[2] | Galo Y. Sara[3] | Martins E. Aisuedion[4] | Kazhigila Thomas [1] | Bashir Hamman[1]

[1]*Department of Chemistry, Modibbo Adama University, Yola, Nigeria*
[2]*Federal College of Horticulture, Dadinkowa, Nigeria*
[3]*Department of Chemistry, Umar Suleiman College of Education, Gashua, Nigeria*
[4]*Engineering Materials Development Institute, Akure, Nigeria*

Use your device to scan and read the article online

Citation J.A. Gidigbi, S.A. Osemeahon, A.B. Abubakar, G.Y. Sara, M.E. Aisuedion, K. Thomas, B. Hamman, **Optimisation of HDPE-PU Thermal Insulation: Effect of Processing Parameters on Insulation Properties.** *Eurasian J. Sci. Technol.*, **2026**, *6*(2), 182-199.

https://doi.org/10.48309/ejst.2026.555764.1280

CC BY

Article info:
Received: 2025-10-27
Accepted: 2026-02-05
Available Online: 2026-05-02
ID: EJST-2510-1280

Keywords:
Thermal Insulation, Energy Conservation, Processing Parameters, Waste Plastics, Copolymerisation

ABSTRACT

Thermal insulation material is crucial for energy maximisation and conservation. The HDPE waste plastics were pyrolysed at 475 °C. The physicochemical parameters analysis such as density, acid value, viscosity, iodine value, pour point, flash point and cloud point were carried out on the HDPE pyrolytic oil (HDPEPO). The physicochemical results agree with literature. The 10% HDPEPO was copolymerised with 90% polyester polyol (PEO) to produce HDPEPO/PEO blend. The blend was used in HDPE-PU thermal insulator production. The effect of processing parameters such as surfactant, catalyst and blowing agent was assessed on important parameters of HDPE-PU thermal insulator. The new formulation conditions established from the optimisation of processing parameters were used to formulate new HDPE-PU thermal insulator and compare with conventional PU thermal insulator and acceptable values for thermal insulatord. The properties of novel HDPE-PU thermal insulator align with and acceptable values, and show better thermal insulation and high compressive strength when compared with conventional PU thermal insulators.

Introduction

Thermal insulation plays a crucial role in energy conservation, building performance, and industrial processes [1]. Chemical processes and process units of an industrial system usually involve heat energy. This heat energy requires adequate management in order for chemical processes to occur. Klemczak *et al*. observed that the concept of thermal insulation transcends beyond industrial applications to numerous applications such as residential and commercial buildings [2]. The importance of thermal insulation has grown significantly due to increasing energy costs, environmental concerns, and stringent building codes aimed at reducing energy consumption [3]. International Energy Agency [4] reported that an effective thermal insulation can reduce energy consumption in buildings by up to 40%, making

*Corresponding Author: gidigbijoshua@gmail.com

it one of the most cost-effective energy conservation measures available.

According to Rodríguez-Soria *et al.* [5] effective thermal insulation materials exhibit low thermal conductivity, typically below 0.1 -0.4 W/m·K; possess air gaps or low-density materials to minimize convective heat transfer and low-emissivity materials to reduce radiative heat transfer in insulation systems [6]. Different materials have been utilized to obtain improved insulation properties. For instance, organic materials such as cellulose derived from recycled paper products, offer thermal conductivity values between 0.035 and 0.040 W/m.K [7]. Mineral materials such as glass wool (fiberglass) derived from melting glass provide thermal conductivity values between 0.032 and 0.045 W/m·K. While glass wool offers combination of better performance, cost-effectiveness, and versatility, its high thermal conductivity value limits its usage as a thermal insulator. Expanded perlite also exhibits insulation properties (0.045-0.065 W/m·K) and finds application in masonry construction and industrial applications [3].

Modern technology has also paved way for the development of new thermal insulation materials such as vacuum insulation panels (VIPs). These materials consist of microporous core material enclosed in a gas-tight envelope under vacuum conditions Kan *et al*. [8]. The VIPs reportedly achieved thermal conductivity values of 0.004-0.008 W/m.K. Nevertheless, high cost of production and potential for vacuum loss limit their applications to specialized uses. Silica aerogel exhibits excellent thermal insulation with very low thermal conductivity. Despite its excellent insulation performance, Koebel *et al*. reported that high costs of production limit widespread adoption [9].

Petrochemical materials such as expanded polystyrene (EPS), extruded polystyrene (XPS), polyurethane, and polyisocyanurate foams are also being deployed as thermal insulators. EPS exhibits thermal conductivity values of 0.030-0.038 W/m.K, while PU thermal insulators demonstrate lower values (0.020-0.025 W/m.K) [10]. The petrochemical thermal insulators offer excellent thermal performance and moisture resistance. They are inexpensive and easy to produce. Nevertheless, the high demand on petrochemical materials for thermal insulator production places huge demand on fossil fuels which is unsustainable and its exploration has detrimental effects on the environment.

Simultaneously, waste plastics remain global epidemic, affecting aquatic life and environment [11]. The alternative use of this waste in chemical processes limit dependence on fossil fuels while reducing waste in the environment, thereby contributing to healthy living and safe environment which are the major cores of global sustainable development goals (SDGs). The previous study has assessed the thermal insulator produced from HDPEPO, including the blending of HDPEPO with conventional PEO and subsequent characterizations. Thus, this study focuses specifically on the optimization of processing parameters (surfactant, catalyst, and blowing agent) in the production of rigid HDPE-PU thermal insulator, aiming to modulate formulation conditions that will be suitable for HDPE-PU thermal insulator, and their corresponding effects of each processing parameter on the important properties of thermal insulator.

Materials and Methods

Calcium carbonate; methylene chloride, magnesium sulfate, flame retardant, distilled water; methylene di-phenyl di-isocyanate (pMDI); PEO; silicone oil; stannous octanoate. All the reagents were of analytical grade. Pyrolyser; weighing balance; infra-red spectrophotometer; stirrer, beakers; and 10 mL measuring cylinder; and stopwatch.

Pyrolysis of HDPE Plastics Waste

The HDPE waste plastics were prepared by firstly washed with water and soap to get rid of oil and dirt (Figure 1). The plastic was cut into pieces and dried at room temperature. The pyrolysis process was carried out at 475 °C.

Physicochemical Properties of HDPEPO

The physicochemical parameters such as density was carried out by Osemeahon *et al*. [12]; pour point, flash point, and cloud point were determined according to the procedure

similar to ASTM (2016). The viscosity and iodine value was measured according to the procedures described by Prurapark *et al*. [13]. The FTIR (Agilent Cary 640) and GC-MS (Agilent 5977C GC/MSD) analyses were subsequently performed on the HDPEPO.

Modification of HDPEPO with PEO

The blending of HDPEPO (0-20%) with PEO was reported in the previous work and subsequent effect of HDPEPO on the crucial parameters of the blends were reported. The properties (such as acid value, hydroxyl value and viscosity) of blend with HDPEPO at 10% align with acceptable values for polyol usually utilized in thermal insulation production.

Formulation of HDPE-PU Thermal Insulator

The formulation of HDPE-PU thermal insulator was carried out according to the method described by Kairytė *et al*. [14] with slight modifications (Table 1). This method required two ways shot where formulation was done in a reacting vessel before transferred to designated mold for casting.

Table 1: Chemicals used in the formulation of rigid HDPE polyurethane thermal insulator

Components	Quantity (g)
HDPEPO	5.00
Silicone oil (surfactant)	0.10
Distilled water (primary blowing agent)	0.20
Stannous octanoate (catalyst)	0.10
pMDI	6.00
Calcium carbonate (Filler)	1.00

Optimisation of the HDPE-PU Thermal Insulator by Studying the Process Parameter

Effect of Surfactant on the Rigid HDPE-Polyurethane Foam Performance

The surfactant (silicone oil) was optimized in the production of rigid HDPE-Polyurethane foam by varying the concentration. Five values of silicone oil 0.0, 0.1, 0.2, 0.3, and 0.4 g were used in the formulation of rigid HDPE-Polyurethane foam.

Effect of the Catalyst on the Rigid HDPE-Polyurethane Foam Performance

The catalyst (Stannous octanoate) was optimised in the formulation of rigid HDPE foam by varying its concentration in the production. The following concentrations 0.0, 0.1, 0.2, 0.3, and 0.4 g were substituted in the formulation of rigid HDPE foam.

Effect of the Blowing Agents on the Rigid HDPE-Polyurethane Foam Performance

The blowing agents were optimised by varying the concentration of blowing agent in the formulation of rigid HDPE-Polyurethane foam. The following concentrations 0.0g, 0.1g, 0.2g, 0.3g and 0.4g were used in rigid HDPE-Polyurethane formulation.

Optimisation of HDPE-PU Thermal Insulator by Studying the Process Parameter

The process parameters such as surfactant (silicone oil); catalyst (stannous octanoate); and blowing agent (water) were optimized by varying each of the parameter's concentration (0.0, 0.1, 0.2, 0.3, and 0.4 g) in the formulation, while keeping others constant.

Result and Discussion

Pyrolysis of HDPE Plastic Waste

The HDPE waste plastics were successfully pyrolyzed to yield HDPEPO with light-brownish color and the char was dark brown. This is also in agreement with the findings of Marwani and Trifarizy, which also reported a light-brownish color for their HDPEPO [15].

Physicochemical Properties of HDPEPO

Table 2 reveals the physicochemical properties of HDPEPO. The acid value of the HDPEPO was lower compared to the findings of Dagde [16], who reported acid value of 0.85 for avocado seed oil biodiesel. The low acid value means that the fuel will not corrode fuel lines or produce sediments.

The flash point of HDPEPO agrees with the findings of Rodriguez Lamar *et al*. for HDPEPO, whereas the density of HDPEPO was lower compared to 0.78 for their HDPE pyrolysis [17]. Pyrolysis temperature determines the density of the oil as lower pyrolysis temperatures produce higher density pyrolysis oil. The viscosity of HDPEPO (1.96) in this study is less than the findings of Prurapark *et al.* [13] which reported 3.297 for their HDPEPO. The significant difference in viscosity may be attributed to the impact of catalyst on the pyrolysis process. The lower viscosity demonstrated by HDPEPO in this study implies its flowability and atomization during engine combustion when used as a fuel.

The iodine value for HDPEPO in this study is similar to the findings of Uebe *et al*. who reported iodine value of 70 for the HDPEPO [18]. Comparing the iodine value with biodiesel, Dagde [16] reported 38.2 for avocado seed oil biodiesel. The high iodine content in HDPEPO implies higher unsaturated compound which can function as a drying oil in coating applications.



Figure 1: Pyrolysis of the HDPE waste plastic to obtain HDPEPO

Table 2: Some physicochemical properties of HDPEPO

Property	HDPEPO
Acid value (mg of KOH/g)	0.57
Flash point (˚C)	35.9
Cloud point (˚C)	10
Pour point (˚C)	-8
Iodine value (g/100)	65
Density (g/cm^3)	0.67
Kinematic viscosity (cSt) at 40˚C	1.96

FTIR Analysis of the HDPEPO

Figure 2 presents the FTIR analysis of the HDPEPO. The FTIR analysis was conducted on the pyrolysis oil obtained from waste HDPE plastic in order to ascertain the functional groups present in the pyrolysis oil. The absorption at the 3,280 cm^{-1} denotes intramolecular alcohol [19]. The detection of hydroxyl group (OH) was also observed in HDPEPO are reported by [20]. The slight differences in OH absorption wavelengths may be attributed to the degradation time for the waste HDPE plastic. Also, absorption at 3,080 cm^{-1} may be attributed to the SP^2 of C-H bond of alkene. This is also supported by the abosorption at 1,649 cm^{-1}. This affirms the presence of olefin (vinylidene) compound in the pyrolysis oil [21]. Moreover, the triplet peaks observed at 2,957; 2,911, and 2,871 cm^{-1} are attributed to sp^3 of C-H bond of alkane. This is also supported by the absorption at 1,455 and 1,377 cm^{-1} which signify C-H bending of alkane [22]. The peak observed at 1,695 cm^{-1} signified the presence of conjugated aldehyde group. Additionally, the peaks observed at 1,151 and 969 cm^{-1} fall within the range of 1,225-950 cm^{-1} which is suggestive of C-H out of plane bending for aromatic compounds [23]. This confirms the presence of aromatic compounds in the waste HDPEPO.

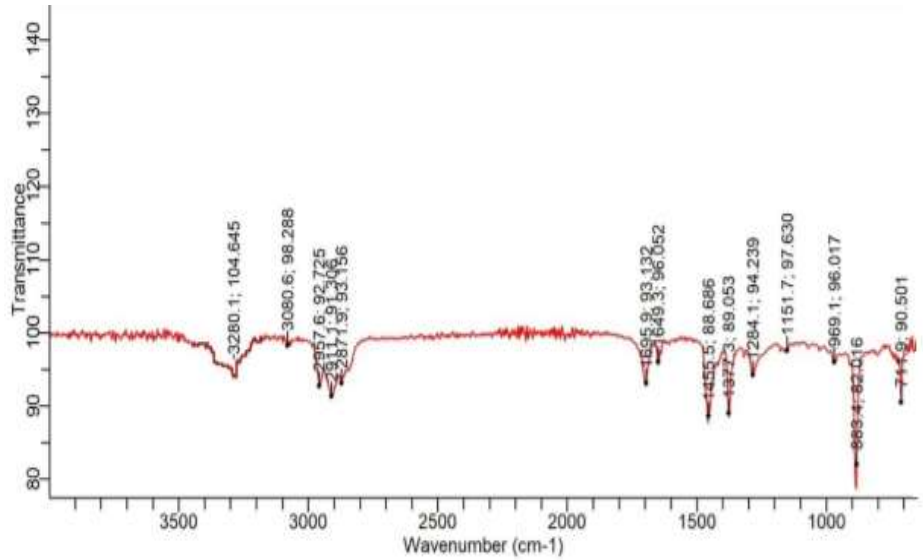


Figure 2: The FTIR of waste HDPEPO

Gas Chromatography-Mass Spectroscopy of HDPEPO

Table 3 presents the major chemical constituents of HDPEPO at 475 °C. The findings revealed that the HDPEPO is predominantly consists of straight-chain alkanes (C_8-C_{20}) and alkenes (1-alkenes), which is consistent with the thermal degradation mechanism of HDPE. The most abundant compounds are tridecane (9.58%), 8-chloro-1-octanol (8.37%), cyclohexanecarboxaldehyde (8.06%) and cyclooctane (7.90%). The smallest fraction of HDPEPO as a light hydrocarbon(C_4-C_8) accounts for 32%, followed by medium hydrocarbons (C_9-C_{16}) at 30%, and heavy hydrocarbons (C_{17}-C_{20+}) which account for the 38% of the total composition. The predominant hydrocarbons (alkenes) in this study agree with the findings of Rodriguez Lamar *et al.* [17], which reported similar hydrocarbon ranges in HDPEPO.

The presence of OH compounds such as 8-chloro-1-octanol ($C_8H_{17}ClO$) and 1-decanol ($C_{10}H_{22}O$) supports the findings of FTIR HDPEPO analysis. This finding is consistent with [20], which reported the presence of hydroxyl compounds during thermal degradation of HDPE waste plastic using pyrolysis method.

Table 3: Major hydrocarbon presents in HDPEPO

Area (%)	Name of compound	Molecular structure
9.01	Tridecane	$C_{13}H_{28}$
8.21	8-Chloro-1-octanol	$C_8H_{17}ClO$
7.89	Cyclohexanecarboxaldehyde	$C_{17}H_{12}O$
7.87	Cyclooctane	C_8H_{16}
6.13	2-Butenoic acid	$C_4H_6O_2$
6.25	Cyclopentane	C_5H_{10}
4.51	Butyl tetratriacontyl ether	$C_{38}H_{78}O$
2.55	2-Hexyl-1-octanol	$C_{14}H_{30}O$
2.53	2-Methyl-E-7-octadecene	$C_{19}H_{38}$
2.23	1-Decanol	$C_{10}H_{22}O$
4.01	3-Hexene	C_6H_{12}
1.95	Dodecane	$C_{12}H_{26}$
1.51	8-Heptadecene	$C_{17}H_{34}$
1.96	Cyclopentane	C_5H_{10}
1.32	1-Decanol	$C_{10}H_{22}O$
1.21	Henicos-1-ene	$C_{21}H_{42}$

Production of HDPE-PU Thermal Insulator

The HDPE-PU thermal insulator was formulated using the procedure stipulated in the experimental design (Figure 3).

FTIR Analysis of HDPE-PU Thermal Insulator Formulated

The absorption at 3,349 cm^{-1} suggestive of hydroxyl (O-H) group from alcohol. The absorption peak near 2,279 cm^{-1} was associated with unreacted NCO from excess pMDI. The dual absorptions observed at 2,972 and 2,868 cm^{-1} are associated with CH_3 and CH_2 chains of the polyol blend. The peak at 1,712 cm^{-1} is suggestive of the C=O of carbonyl in polyurethane structure. The peak at 1,599 cm^{-1} suggestive of CONH band while 1,541 cm^{-1} indicates the presence of aliphatic nitro compounds N-O which might be due to isocyanates [24]. The absorption at 1,658 cm^{-1} suggested of -C=C- from possible aromatic content of HDPEPO. The absorptions at 1,086.5 cm^{-1} and 1,455 cm^{-1} are indicative of C-O and C-N of polyurethane thereby confirming chemical interaction between HDPEPO/PEO and isocyanate [25]. The peak at 1,508 cm^{-1} was observed, which is an indicative of C-H stretching of aryl of isocyanate (Figure 4).



Figure 3: The HDPE-PU thermal insulator

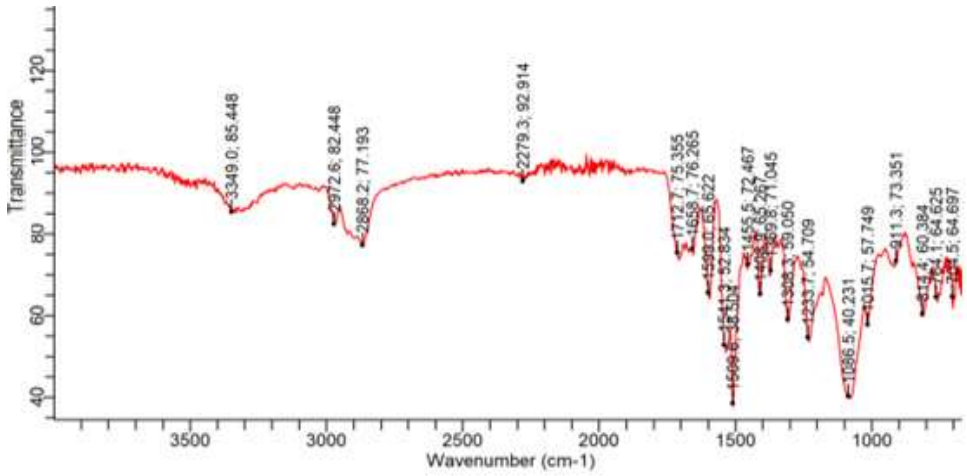


Figure 4: FTIR of theHDPE-PU thermal insulator

Comparison of Physicochemical Properties of Rigid HDPE-Polyurethane Foam and Acceptable Value in Foam Industries

Table 4 presents the comparison of physicochemical properties of rigid HDPE-polyurethane foam produced from HDPEPO/PEO blend and the acceptable values for the rigid polyurethane foam. The forming parameters such as cream time, gel time, and tack-free time showed that they are within the generally acceptable range for rigid polyurethane foam with the exception of gel time. The low gel time was also noted by Suleiman *et al*. [26]. The low gel time may be attributed to moderate hydroxyl value of the HDPEPO/PEO blend, as the hydroxyl groups available quickly cross-linked with isocyanate to form urethane linkages [27].

Additionally, gel time is also dependent on the quantity of catalyst used in the formulation. However, the density of the rigid HDPE-polyurethane foam produced fit into general acceptable range, while failing to meet the optimal value for structural applications. The reason may be attributed to the usage of surfactants, which may affect the mixture of all the chemical componenets in the formulation. The water absorptivity of the HDPE-polyurethane foam was still within the acceptable level, while compressive strength still in agreement with the general acceptable value. The thermal conductivity of the rigid HDPE-polyurethane foam observed to be within the acceptable level.

The closed cell content was determined by assessing SEM image of the foam cell [27]. The HDPE-polyurethane foam cells were observed to be closed with distinct cell walls [28].

Table 4: Comparison of physicochemical properties of HDPE-PU thermal insulator with acceptable value for conventional thermal insulator

Parameters	HDPE-PU thermal insulator	Acceptable value [29-31]
Cream time (s)	9 ±0.02	5-15
Gel time(s)	17 ±0.03	30-120
Tack-free time (s)	63 ±0.01	30-120
Density (g/cm^3)	0.032 ±0.03	0.03-0.1
Water absorptivity (%)	1.1±0.04	<2
Compressive strength (kPa)	120 ±0.02	100-400
Thermal conductivity (W/m·K)	0.023 ±0.03	0.02-0.03

Experiments are done in triplicate with standard deviation.

Effect of Optimisation of Process Parameter on the Cream Time of HDPE-PU Thermal Insulator

Figure 5 reveals the effect of the optimization of process parameters on the cream time of HDPE-PU thermal insulator. The increase in concentration of surfactant and catalyst from 0.0 to 0.1 concentration (g) sharply decreases the cream time of HDPE-PU thermal insulator, while opposite was observed for blowing agent. The sharp decline due to the surfactant is as a result of lack/poor chemical interface interaction of the foam components caused by the existence of high surface tension between the foam components. However, as the concentration of surfactant increases, surface tension is reduced and gas-liquid interface interaction is promoted, thus facilitating bubble formation and faster nucleation. The sharp decline due to the catalyst is a result of delay in gelation process. However, as the concentration of catalyst increases, polymerisation reaction between polyol (HDPEPO/PEO) and diisocyanate is promoted, thereby facilitating a faster foaming process. This finding is similar to Kiss *et al.* , who reported that increase in tertiary amine catalysts reduces cream time [32].

The sharp increase was as a result of low diisocynate-H_2O reaction to evolve gas; hence, the cream time prolonged. Conversely, the decrease in cream time observed was as a result of the acceleration of initial foam rise as water reacts with isocyanate to generate CO_2. This result is similar to the report of Al-Moameri *et al.* [33], who also emphasized that cream time decreases with an increase in blowing agent.

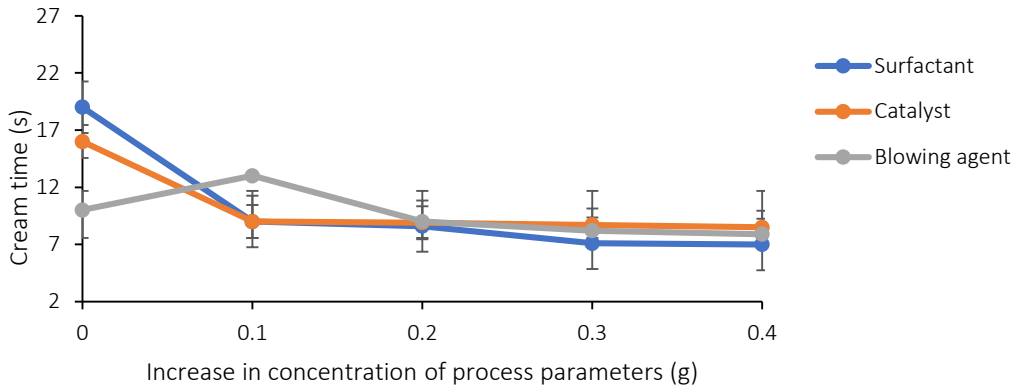


Figure 5: Effect of increase in concentration of process parameter on the cream time of HDPE-PU thermal insulator

Effect of Optimisation of Process Parameter on the Gel Time of HDPE-PU Thermal Insulator

Figure 6 shows the impact of optimizing the process parameters on the gel time of HDPE-PU thermal insulator. The gel time decreases sharply as the concentration (surfactant and catalyst) increases from 0.0 to 0.1. The sharp decline as a result of the surfactant is due to the improved interface between polyol and diisocyanate. In addition, a steady decrease was observed as the interface between polyol (HDPEPO/PEO) and diisocyanate increases. At 0.4 g of surfactant, seems the surfactant reaches its peak as there is no significant difference between gel time at 0.3 and 0.4 g. This may be due to complete polymerization between the polyol and diisocyanate. Zhang *et al.* [34] demonstrated that increasing surfactant concentration from 1 to 3 parts per hundred polyol (pphp) decreased gel time by approximately 15-20%.

The decrease in gel time as a result of increased catalyst concentration is due to the improved rate of polymerization reactions, thereby aiding crosslinking network formation. This outcome agrees with Altuna *et al.* [35] who demonstrated that an increase in amine catalyst concentration reduced gel time of epoxy vitrimers. Meanwhile, there was a steady increase in gel time as the blowing agent increased. This behavior is attributed to reduction in reaction temperature

by absorbing heat thereby slowing down the polymerization (isocyanate-polyol) process. This finding agrees Shalbafan *et al.* [36], with which also emphasized that blowing agent affects the degree of polymerization of PU conventional thermal insulator.

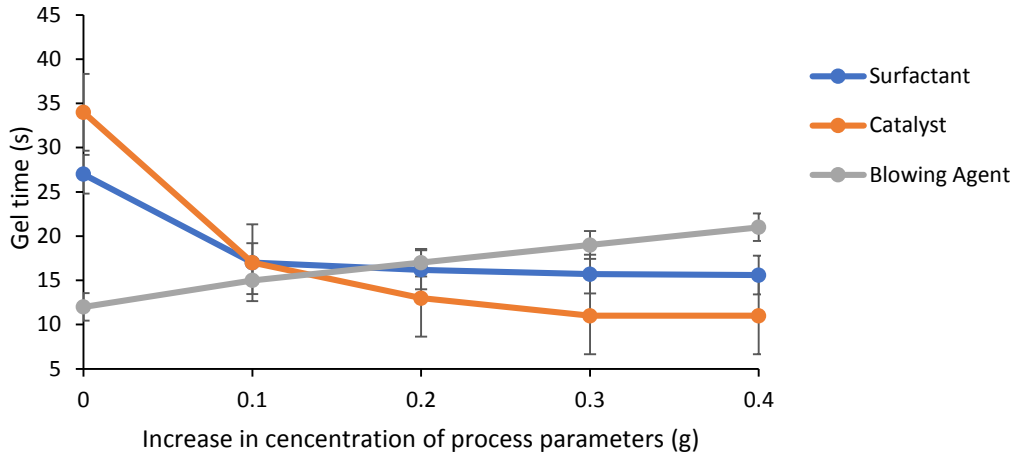


Figure 6: Effect of optimisation of process parameter on the gel time of HDPE-PU thermal insulator

Effect of Optimization of Process Parameters on the Tack-Free Time of HDPE-PU Thermal Insulator

Figure 7 presents the result of optimization of the process parameters on the tack-free time of HDPE-PU thermal insulator. The sharp decline observed in tack-free time as the concentration of both surfactant and catalyst increases was due to the promotion of more uniform cell structure and faster surface curing [14]. This outcome agrees with Kim *et al*. [37], which reported an almost 30% decrease in tack-free time of rigid PIR foam, as the concentration of silicone surfactant increases from from 1 to 2.5 pphp. While the decrease in gel time as the concentration of the catalyst increases was as a result of improved cross-linking, since the catalyst enhances faster crosslinking reactions. Ivdre *et al.* [27] observed that increasing tin catalyst concentration reduced tack-free time of the PU foam. Meanwhile, the tack-free time increases as the concentration of the blowing agents increases. This due to the increased gas evolution (CO_2 gas) from diisocyanate and water reaction. Shalbafan *et al* [36] observed similar findings and attributed to gas evolution and temperature reduction effects on the reaction.

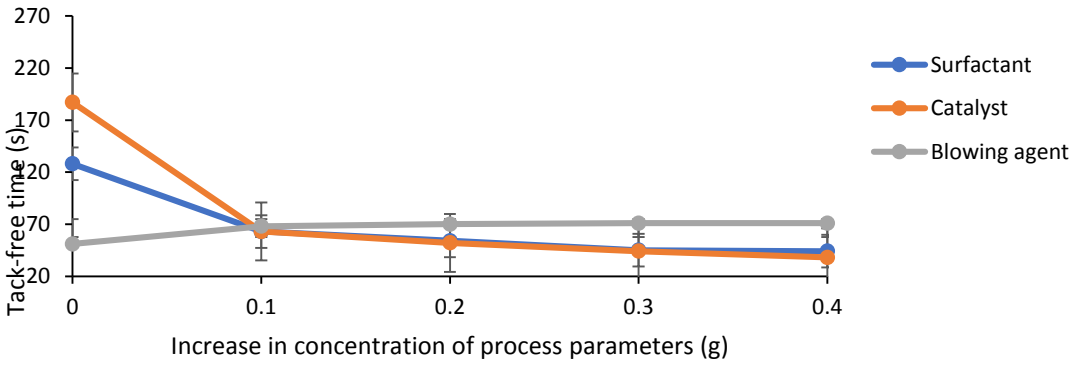


Figure 7: Effect of optimisation of process parameter on the tack-free time of HDPE-PU thermal insulator

Effect of Optimization of Process Parameter on the Density of HDPE-PU Thermal Insulator

Figure 8 shows the optimzsation of process parameters on the density of HDPE-PU thermal insulator. The density of rigid HDPE-PU increases as the concentration of surfactant and catalyst increases. The increase in density due to surfactant was attributed to the stabilization effect of surfactant on the polyol-isocyanate (crosslinking) reaction, while that of catalyst was as a result of the high degree of crosslinking between polyol and isocyanate to achieve complete reaction. This implies that the much of the hydroxyl group is successfully reacted with cyanate (OCN) from isocyanate to form HDPE-PU thermal insulator. However, it is worth noting that the reaction could not take place at a 0.0 concentration of catalyst due to energy barrier between polyol and diisocyanate.

Meanwhile, the density rigid HDPE-PU decreases with increases in blowing agent. This is due to the fact that more gases from the bubble formation create additional cells, expanding the foam volume while maintaining the same mass of polymer [38]. Moreover, Acquah = reported that increasing water content from 2 to 5 pphp reduced foam density from 45 kg/m^3 to 28 kg/m^3 [39].

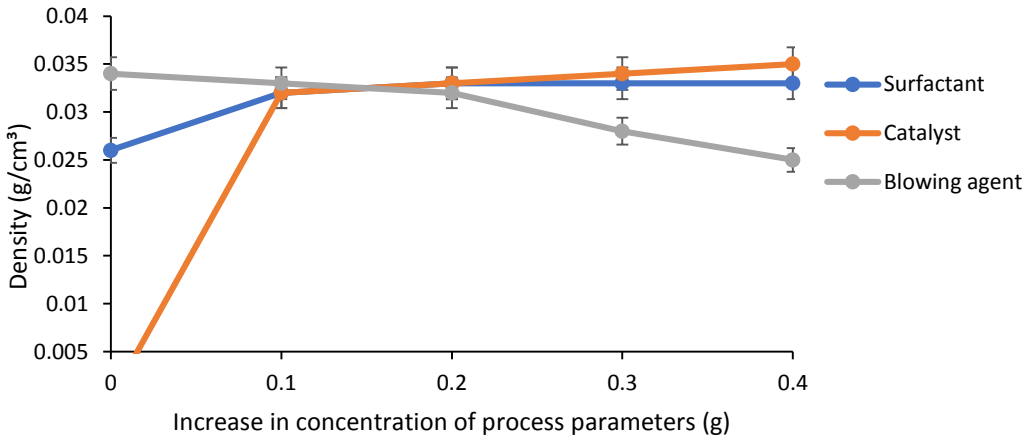


Figure 8: Effect of optimisation of process parameter on the density of HDPE-PU thermal insulator

Effect of Optimization of Process Parameters on the Water Absorptivity of HDPE-PU Thermal Insulator

Figure 9 depicts the effect of increasing in process parameters on the water absorptivity of the HDPE-PU thermal insulator. The increase in surfactant significantly increases the water absorptivity as a result of irregular cell formation, and decreases steadily due to the formation of high cross-linked uniform cell structures, which may restrict the passage of water into PU foam matrix. 0.40 g of surfactant seems to be the peak due to very slight increment observed. Any further increase from this point may lead to excess, which may affect water absorptivity.

Zhang *et al.* [40] reported that excessive surfactant may increase water absorption due to an increase in percentage of open cells. The water absorptivity of the HDPE-PU thermal insulator increases as the catalyst concentration increases. This is due to the fact that catalyst promotes more complete crosslinking network, thereby creating a more closed-cell structure. However, Shalbafan *et al.* [36] observed that addition of excess catalyst beyond the saturated concentration may lead to irregular cell structures with potential cell rupture, potentially increasing water absorption.

Meanwhile, the water absorptivity increases as the concentration of blowing agent increases. This is due to the formation of thinner cell walls and potentially more open cells, creating

pathways for water penetration. This finding aligns with Berardi [41] who observed that water absorption of PU foam increasing from 2.5% to 4.8% when blowing agent concentration was doubled.

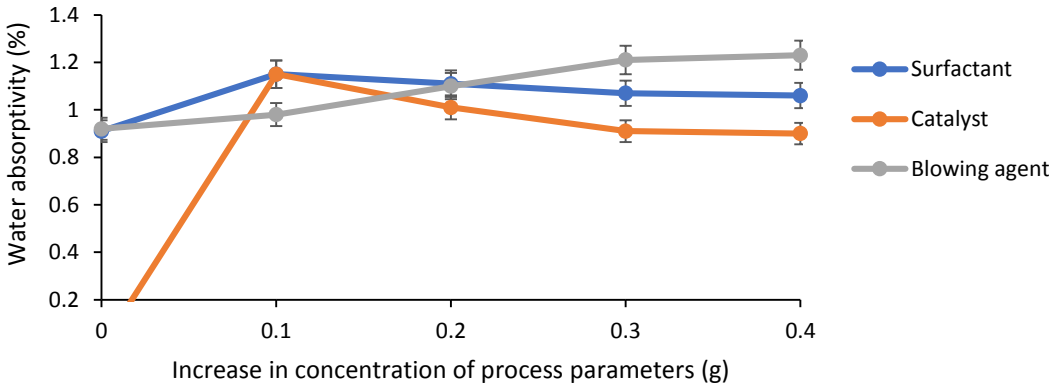


Figure 9: Effect of optimization of process parameter on the water absorptivity of HDPE-PU thermal insulator

Effect of Optimization of Process Parameter on the Compressive Strength of HDPE-PU Thermal Insulator

According to Figure 10, the increase in catalyst concentration in HDPE-PU thermal insulator formulation improves compressive strength by creating more uniform cell structures and higher crosslinking density. This finding agrees with Zhang *et al*. [34], who reported that catalysts aid the formation of uniform cells, which enhance resistance against stress. However, excess catalyst may lead to irregular cell formation and potential stress concentration points which may reduce compressive strength of the foam. The increase in surfactant concentration gives rise to increase in compressive strength. This is due to the fact that increase in surfactant produces more uniform cell structures with thicker cell walls. This finding agrees with Majida *et al.* [42] who reported that compressive strength increased when surfactant concentration increased in rigid PU/Kenaf fiber. However, there is a slight decrease in the compressive strength at 0.4 g of surfactant. This may be as a result of creation of very small cells with thin cell walls.

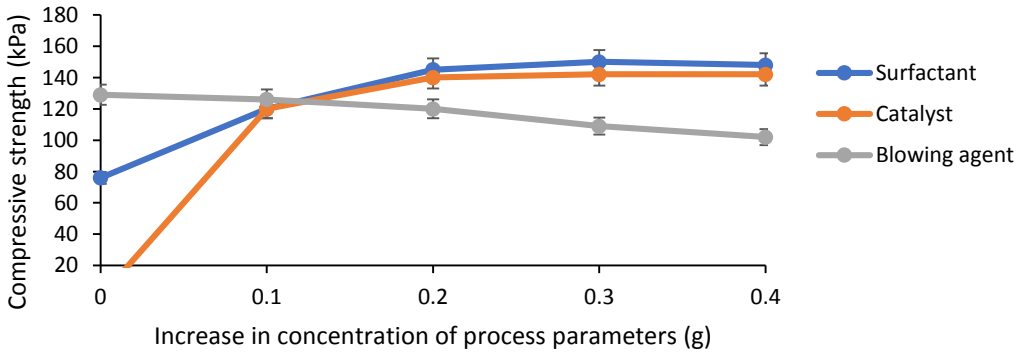


Figure 10: Effect of optimization of process parameter on the compressive strength of HDPE-PU thermal insulator

Kairytė *et al*. [14] observed that addition of excess surfactant could lead to lesser compression strength due destabilization the foam structure caused by excess surfactant. The compressive strength decreases as the concentration of blowing agent increases. This is

because blowing agent creates an expanded volume of foam with less material per unit volume to resist compression [43]. This finding aligns with Li *et al.* [44], who reported that an increase in the water content decreases the compressive strength of PU foam.

Effect of Optimization of Process Parameter on the Thermal Conductivity of HDPE-PU Thermal Insulator

Figure 11 presents the effect of increasing in process parameters on the thermal conductivity of the HDPE-PU thermal insulator. The thermal conductivity of the HDPE-PU thermal insulator decreases with an increase in concentration of surfactant. This is due to the formation of more uniform closed cell structures with better capacity for insulation, as the surfactant aids better miscibility of foam's reagents. The thermal conductivity at 0.4 g of surfactant remains the same with 0.3 g. This may be the value at which highest miscibility of reagents was achieved, and any further miscibility may affect the structural matrix of the foam. Berardi [40] reported that higher surfactant concentrations could lead to cell window rupture thereby, causing foam with high thermal conductivity and lesser insulative properties.

The thermal conductivity behavior in relation to the increase in catalyst concentration was due to the fact that increase in catalyst concentration enhanced the reactivity of monomers which resulted in smaller, more uniform closed cells that trap insulating gas more effectively, leading to better insulation performance. This finding aligns with Cunningham *et al.* [45], who observed a decrease in thermal conductivity as the catalyst concentration increased moderately. The thermal conductivity of HDPE-PU thermal insulator decreases with increases in concentration of blowing agent. This is due to high CO_2 generated from water-isocyanate reactions which create more open cells. The more cells generated, the thinner the cell wall, making it susceptible to heat conductivity [25].

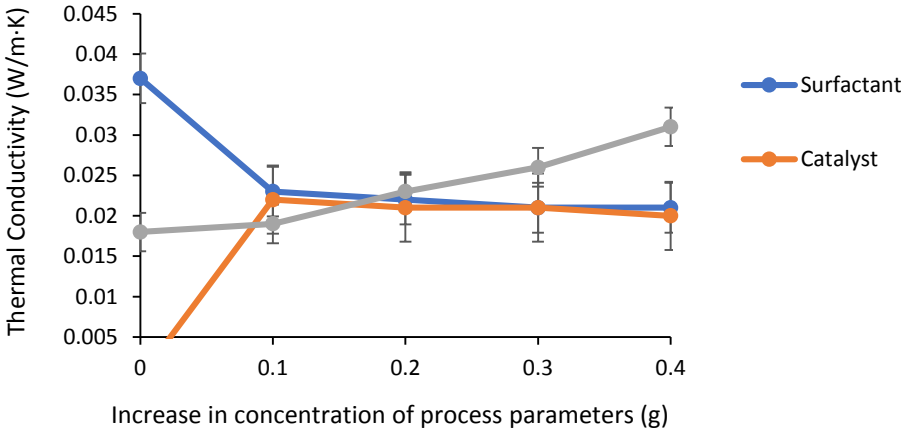


Figure 11: Effect of optimization of process parameter on the thermal conductivity of HDPE-PU thermal insulator

Formulation of HDPE-PU Thermal Insulator from the Optimized Parameters

The rigid HDPE-polyurethane foam was formulated using the optimised processing parameters. The formulation was done according to Table 5. The formulated rigid HDPE-polyurethane thermal insulator was characterized in order to compare with the acceptable value.

Table 5: The formulation for HDPE-PU thermal insulator

Components	Quantity (g)
HDPEPO/PEO	5.00
Silicone oil (surfactant)	0.20
Distilled water (primary blowing agent)	0.20
Stannous octanoate (catalyst)	0.20
pMDI	5.00
Calcium carbonate (Filler)	1.00



Figure 12: The HDPE-PU thermal insulator produced from the optimised formulation condition

Characterisation of the Optimized HDPE-PU Thermal Insulator

The HDPE-PU thermal insulator formulated using the novel optimised formulation (Figure 12) and was characterise to compare with the standard. Table 6 reveals the properties of novel HDPE-PU thermal insulator and comparison with the standard.

Cream Time

The HDPE-PU thermal insulator exhibits superior processing characteristics with a reduced cream time of 9±0.02 seconds compared to 13±0.05 seconds for conventional rigid PU thermal insulator. The reduced cream time implies enhanced reactivity due to catalyst efficiency in the rigid HDPE-PU modified matrix. This influence the leverage on density and compressive strength of the HDPE-PU thermal insulator. The cream time is still within the acceptable limit in the rigid PU production. The cream time of HDPE-PU thermal insulator in this study is lower when compared with the 25s reported for PU thermal insulators [46]. This difference in cream time is associated with the higher functional polyol since polyether polyol was used in that study.

Gel Time

The HDPE-PU thermal insulator (17 ± 0.04) has a longer gel time when compared with conventional PU thermal insulator (15 ± 0.03). This increase in gel time may be as a result of the presence of hydrophobic HDPEPO. The longer gel time allows for controlled polymerization with steady curing process. The gel time of both rigid PU and HDPE-PU thermal insulators does not align with the acceptable level. This may be due to the low hydroxyl polyol utilized in the formulation, as higher hydroxyl value requires longer time for polyol-isocyanate reaction to complete [47].

Tack-Free Time

The tack-free time of rigid HDPE-PU was observed to be 68 ± 0.04 as against 66 ± 0.02 seconds reported for conventional PU thermal insulator. The differences in both tack-free times may be due to the fact that PU thermal insulator exhibits faster surface curing, as a result of fast crosslinking process, whereas, the organic compounds in HDPEPO/PEO polyol may slow down its curing. According to Nisa *et al* [48], a moderate curing results in the production of finer closed-cell PU thermal insulators. Additionally, Ivdre *et al*. [27] reported cream time of 25 seconds; gel time of 56 seconds and

tack-free time of 108 seconds for rigid polyurethane foam produced from RO/PET.

Density

The HDPE-PU thermal insulator demonstrates higher density of 0.047 ± 0.03 g/cm^3, when compared with 0.033 ± 0.07 g/cm^3 observed for conventional PU thermal insulator. The higher density is as a result of contribution of cyclic and aliphatic compound present in the HDPEPO/PEO blend which does not undergo total polymerisation. This density increase is accompanied by improved property consistency, as evidenced by the reduced standard deviation (±0.03 vs. ±0.07). Chaudhary *et al*., [49] reported that bio-based polyurethane foams have densities in the range of 23-41 kg/m^3 for thermal insulation applications, positioning the HDPE-PU thermal insulator at 47 kg/m^3 as competitive for enhanced mechanical performance applications. The controlled density increase suggests that HDPE incorporation creates a more uniform cell structure while providing reinforcement, which is critical for long-term dimensional stability and mechanical performance [50].

Water Absorptivity

The HDPE-PU thermal insulator demonstrates a good water resistance capability (0.9 ± 0.01%) when compared to conventional PU thermal insulator (1.2 ± 0.01%). This reduction in water absorptivity is associated with the inherent hydrophobic nature of HDPEPO in HDPEPO/PEO polyol. This reduction in water absorption is particularly significant for thermal insulation applications, as moisture uptake can severely compromise insulating performance over time. The result is close to the findings of Jabar *et al*. [51] which reported 0.95% for rigid polyurethane foam formulated from vegetable polyols.

Compressive Strength

The compressive strength of HDPE-PU thermal insulator shows significant improvement from 157 ± 0.03 kPa to 121 ± 0.02 kPa observed in conventional PU thermal insulator. This may be attributed to HDPE's reinforcing effect on the polyurethane matrix, creating a composite structure that better distributes applied loads and resists deformation. This improvement is remarkable considering that both values fall well within the acceptable range of 100-400 kPa. He *et al.,* [52] on polyurethane foams reported compressive strengths ranging from 45-188 kPa for thermal insulation applications, thereby positioning them in the upper performance range.

Thermal Conductivity

The HDPE-PU thermal insulator demonstrates lower conductivity with 0.021 ± 0.06 W/m·K compared to 0.025 ± 0.07 W/m·K for conventional PU thermal insulator. This implies that the lower the thermal conductivity correlates the higher the insulation performance. The most significant improvement is observed in thermal conductivity, with the HDPE-PU thermal insulator. This 16% reduction in thermal conductivity represents a substantial improvement in insulating performance. The HDPE-PU thermal insulator has a better insulation properties when compared with PU thermal insulator rigid polyurethane foam modified with fly ash of 0.022-0.028 W/mK thermal conductivity [53]. However, Carriço *et al*. [54] reported that bio-based foams show higher values (0.031-0.063 W/m.K). Comparison of optimized HDPE-PU thermal insulator properties with other thermal insulator properties was presented in Table 7.

Table 6: Properties of optimisedHDPE-PU thermal insulator and standard for thermal Insulation application

Parameters	Rigid PU foam	HDPE-PU thermal insulator	Acceptable value [29-31]
Cream time (s)	13±0.05	9±0.02	5-15
Gel time(s)	15±0.03	17 ± 0.04	30-120
Tack-free time (s)	66 ±0.02	68 ± 0.04	30-120

Density (g/cm^3)	0.033 ±0.07	0.047+ 0.03	0.03-0.1
Water absorptivity (%)	1.2±0.01	0.9 ±0.01	<2
Compressive strength (kPa)	121 ± 0.02	157 ±0.03	100-400
Thermal Conductivity (W/m.K)	0.025±0.07	0.021±0.06	0.02-0.03

Table 7: Comparison of some physical properties of HDPE-PU thermal insulator with literature

Physical property								
Type of thermal Insulator	**Cream time (s)**	**Gel time (s)**	**Tack-free time (s)**	**Density (g/cm^3)**	**Water absorptivity (%)**	**Compressive strength (kPa)**	**Thermal conductivity (W/m·K)**	**Literature**
HDPE-PU	9.02	17.01	68.02	0.047	0.9	157	0.021	This study
PBRPU	NR	NR	NR	0.040	0.95	280	NR	[51]
SOY_PU	7	15	60	0.039	NR	NR	NR	[26]
RO/PET-PU	22	52	75	0.039	2.9	450	0.021	[27]
LBP-PU	NR	NR	NR	0.021	NR	NR	0.038	[28]
LP_PU	NR	NR	NR	0.120	NR	40	0.070	[25]
KF-PU	NR	NR	NR	NR	NR	712.8	NR	[42]

**NR means not reported.*

Conclusion

The study assessed the optimization of HDPE-PU thermal insulator by varying the concentration of processing parameters such as surfactant, catalyst and blowing agent. The effect of these processing parameters was assessed on crucial properties of HDPE-PU thermal insulator. Increases in surfactant concentration contribute positively to density, water absorptivity, compressive strength and thermal conductivity, while blowing agent affects these properties. Additionally, the novel formulation conditions revealed an improved HDPE-PU thermal insulator. Moreover, the optimized HDPE-PU thermal insulator performs better when compared with conventional thermal insulators. Furthermore, the properties of HDPE-PU thermal insulator demonstrate improvement when compared with other properties of thermal insulators. Hence, HDPE-PU thermal insulator is efficient, low-cost and sustainable.

ORCID

Joshua A. Gidigbi
https://orcid.org/0000-0001-7981-4792
Sunday A. Osemeahon
https://orcid.org/0000-0002-7691-5763
Abdulrasaq B. Abubakar
https://orcid.org/0000-0002-9842-4551

References

[1] Ali, A., Issa, A., Elshaer, A. A comprehensive review and recent trends in thermal insulation materials for energy conservation in buildings. *Sustainability*, **2024**, 16(20), 8782.
[2] Klemczak, B., Kucharczyk-Brus, B., Sulimowska, A., Radziewicz-Winnicki, R. Historical

evolution and current developments in building thermal insulation materials—A review. *Energies*, **2024**, 17(22), 5535.
[3] Kumar, A., Suman, B. Experimental evaluation of insulation materials for walls and roofs and their impact on indoor thermal comfort under composite climate. *Building and environment*, **2013**, 59, 635-643.
[4] Ibrahim, I.A., Ötvös, T., Ghanem, C., Rocca, E., Gilmanova, A. International Energy Agency (IEA), **2021**.
[5] Rodríguez-Soria, B., Domínguez-Hernández, J., Pérez-Bella, J.M., del Coz-Díaz, J.J. Review of international regulations governing the thermal insulation requirements of residential buildings and the harmonization of envelope energy loss. *Renewable and Sustainable Energy Reviews*, **2014**, 34, 78-90.
[6] Tychanicz-Kwiecień, M., Wilk, J., Gil, P. Review of high-temperature thermal insulation materials. *Journal of Thermophysics and heat Transfer*, **2019**, 33(1), 271-284.
[7] De Zoysa, R.N. Kristombu Baduge, K.S. Thilakarathna, P.S.M. Liu, Xuemei Costa, S. Gunarathne, U. Cazacu, E. Braunsch, T. Eco-friendly materials for structural insulated panels: A comprehensive review. *Journal of Building Engineering*, **2025**, 113(1), 114059.
[8] Kan, A., Zhang, X., Chen, Z., Cao, D. Effective thermal conductivity of vacuum insulation panels prepared with recyclable fibrous cotton core. *International Journal of Thermal Sciences*, **2023**, 187, 108176.
[9] Koebel, M., Rigacci, A., Achard, P. Aerogel-based thermal superinsulation: an overview. *Journal of Sol-Gel Science and Technology*, **2012**, 63(3), 315-339.
[10] Respondek, Z. Possibilities of improving the thermal protection of residential buildings by the application of over-rafter roof insulation systems. *Quality Production Improvement-QPI*, **2021**, 3(1), 283-292.
[11] Alabi, O.A., Ologbonjaye, K.I., Awosolu, O., Alalade, O.E. Public and environmental health effects of plastic wastes disposal: a review. *J Toxicol Risk Assess*, **2019**, 5(021), 1-13.
[12] Osemeahon, S.A., Dass, P.M., Aliyu, B.A., Fadawa, F.G. Development of an Emulsion Paint Binder from the Copolymerization of Dimethylol Urea and Hydroxylated Mahogany Seed Oil. *International journal of recent innovations in academic research*, **2018**, 2(7), 26-39.
[13] Prurapark, R., Owjaraen, K., Saengphrom, B., Limthongtip, I., Tongam, N. Effect of temperature on pyrolysis oil using high-density polyethylene and polyethylene terephthalate sources from mobile pyrolysis plant. *Frontiers in Energy Research*, **2020**, 8, 541535.
[14] Kairytė, A., Kremensas, A., Balčiūnas, G., Członka, S., Strąkowska, A. Closed cell rigid polyurethane foams based on low functionality polyols: Research of dimensional stability and standardised performance properties. *Materials*, **2020**, 13(6), 1438.
[15] Marwani, M., Trifarizy, M.D. the Pyrolysis of High-Density Polyethylene (Hdpe) and Polypropylene (Pp) Plastic Waste Blend. *Indonesian Journal of Engineering and Science*, **2024**, 5(1), 029-034.
[16] Dagde, K.K. Extraction of vegetable oil from avocado seeds for production of biodiesel. *Journal of Applied Sciences and Environmental Management*, **2019**, 23(2), 215-221.
[17] Rodriguez Lamar, Y., Noboa, J., Torres Miranda, A.S., Almeida Streitwieser, D. Conversion of PP, HDPE and LDPE plastics into liquid fuels and chemical precursors by thermal cracking. *Journal of Polymers and the Environment*, **2021**, 29(12), 3842-3853.
[18] Uebe, J., Lekaviciute, E., Kryzevicius, Z., Zukauskaite, A. Comparison of Antioxidants to Increase the Oxidation Stability of Pyrolysis Oils of Three Plastics Using Iodine Value. *Processes*, **2024**, 12(4), 638.
[19] Li, J., Wang, J., Liu, J., Li, Y., Ma, H., Yang, J., Zhang, Q. Facile synthesis of multi-type carbon doped and modified nano-TiO2 for enhanced visible-light photocatalysis. *RSC advances*, **2020**, 10(70), 43193-43203.
[20] Liandi, A.R., Solehah, M.A., Setyono, M.H., Aziz, I., Siregar, Y.D.I. Pyrolysis of PP and HDPE from plastic packaging waste into liquid hydrocarbons using natural zeolite Lampung as a catalyst. *Case Studies in Chemical and Environmental Engineering*, **2023**, 7, 100290.
[21] Gin, A.W., Hassan, H., Ahmad, M.A., Hameed, B. H., Din, A.M. Recent progress on catalytic co-pyrolysis of plastic waste and lignocellulosic biomass to liquid fuel: The influence of technical and reaction kinetic parameters. *Arabian Journal of Chemistry*, **2021**, 14(4), 103035.
[22] Alawa, B., Chakma, S. Synergism and production of hydrocarbon-rich fuel from mixed-feedstock through co-pyrolysis of LDPE and PP: an assessment of fuel properties, engine performance,

and gas emission. *Journal of Analytical and Applied Pyrolysis*, **2022**, 168, 105736.
[23] Singh, R.K., Ruj, B., Sadhukhan, A.K., Gupta, P. A TG-FTIR investigation on the co-pyrolysis of the waste HDPE, PP, PS and PET under high heating conditions. *Journal of the Energy Institute*, **2020**, 93(3), 1020-1035.
[24] Mahmood, N., Yuan, Z., Schmidt, J., Xu, C.C. Preparation of bio-based rigid polyurethane foam using hydrolytically depolymerized Kraft lignin via direct replacement or oxypropylation. *European Polymer Journal*, **2015**, 68, 1-9.
[25] Wu, M., Peng, J.J., Dong, Y.M., Pang, J.H., Zhang, X.M. Preparation of rigid polyurethane foam from lignopolyol obtained through mild oxypropylation. *RSC Advances*, **2022**, 12(34), 21736-21741.
[26] Suleiman, A.S., Dukhan, N. Long-domain simulation of flow in open-cell mesoporous metal foam and direct comparison to experiment. Microporous and mesoporous materials, **2014**, 196, 104-114.
[27] Ivdre, A., Abolins, A., Sevastyanova, I., Kirpluks, M., Cabulis, U., Merijs-Meri, R. Rigid polyurethane foams with various isocyanate indices based on polyols from rapeseed oil and waste PET. *Polymers*, **2020**, 12(4), 738.
[28] Vieira, F.R., Gama, N.V., Barros-Timmons, A., Evtuguin, D.V., Pinto, P.C. Development of rigid polyurethane foams based on Kraft lignin polyol obtained by oxyalkylation using propylene carbonate. *ChemEngineering*, **2022**, *6*(6), 95.
[29] EN 826. Thermal insulating products for building applications – Determination of compression behavior. *European Committee for Standardization*, **2013**.
[30] Kabakci, E., Sayer, G., Suvaci, E., Uysal, O., Güler, İ., Kaya, M. Processing-structure-property relationship in rigid polyurethane foams. *Journal of Applied Polymer Science*, **2017**, 134(21).
[31] Petcu, C., Hegyi, A., Stoian, V., Dragomir, C.S., Ciobanu, A.A., Lăzărescu, A.V., Florean, C. Research on thermal insulation performance and impact on indoor air quality of cellulose-based thermal insulation materials. *Materials*, **2023**, 16(15), 5458.
[32] Kiss, G., Rusu, G., Bandur, G., Hulka, I., Romecki, D., Péter, F. Advances in low-density flexible polyurethane foams by optimized incorporation of high amount of recycled polyol. *Polymers*, **2021**, 13(11), 1736.
[33] Al-Moameri, H.H., Nabhan, B.J., M, T.W., Abdulrehman, M.A. Impact of blowing agent-blends on polyurethane foams thermal and mechanical properties. In *AIP Conference Proceedings*, **2020**, 2213, 20177.
[34] Zhang, G., Zhang, Q., Wu, Y., Zhang, H., Cao, J., Han, D. Effect of auxiliary blowing agents on properties of rigid polyurethane foams based on liquefied products from peanut shell. *Journal of Applied Polymer Science*, **2017**, 134(48), 45582.
[35] Altuna, F.I., Hoppe, C.E., Williams, R.J. Epoxy vitrimers with a covalently bonded tertiary amine as catalyst of the transesterification reaction. *European Polymer Journal*, **2019**, 113, 297-304.
[36] Shalbafan, A., Choupani Chaydarreh, K., Welling, J. Effect of blowing agent concentration on rigid polyurethane foam and the properties of foam-core particleboard. *Wood Material Science & Engineering*, **2021**, 16(2), 85-93.
[37] Kim, H.J., Jin, X., Choi, J.W. Investigation of bio-based rigid polyurethane foams synthesized with lignin and castor oil. *Scientific Reports*, **2024**, 14(1), 13490.
[38] Dong, M., Wang, G., Zhang, X., Tan, D., Ren, J., Colorado, H., Guo, Z. An overview of polymer foaming assisted by supercritical fluid. *Advanced Composites and Hybrid Materials*, **2023**, 6(6), 207.
[39] Acquah, E.K. Synthesis of Liquid Lignin Polyol Designed for Flexible Polyurethane Foams. *Doctoral Dissertation, Michigan State University*, **2025**.
[40] Zhang, G., Wu, Y., Chen, W., Han, D., Lin, X., Xu, G., Zhang, Q. Open-cell rigid polyurethane foams from peanut shell-derived polyols prepared under different post-processing conditions. *Polymers*, **2019**, 11(9), 1392.
[41] Berardi, U. The impact of aging and environmental conditions on the effective thermal conductivity of several foam materials. *Energy*, **2019**, 182, 777-794.
[42] Majida, R.A., Akhirb, F.S.M., Ghazalib, S.K., Mohamadb, Z. Thermal stability and compression strength of rigid polyurethane/kenaf fibre biocomposite foam. *Chemical Engineering*, **2020**, 78, 22-34.
[43] Bisht, A., Gangil, B., Patel, V.K. Selection of blowing agent for metal foam production: A review. *Journal of Metals, Materials and Minerals*, **2020**, 30(1), 1-10.
[44] Li, M., Du, M., Wang, F., Xue, B., Zhang, C., Fang, H. Study on the mechanical properties of polyurethane (PU) grouting material of different geometric sizes under uniaxial compression. *Construction and Building Materials*, **2020**, 259, 119797.

[45] Cunningham, V.J., Giakoumatos, E.C., Marks, M., Armes, S.P., Wanless, E.J. Effect of morphology on interactions between nanoparticle-stabilised air bubbles and oil droplets. *Soft Matter*, **2018**, 14(17), 3246-3253.

[46] Rifathin, A., Dara, F., Novriadi, D., Radini, F.A., Vicarneltor, D.N., Burhani, D., Sondari, D., Septevani, A.A. Study on properties of rigid polyurethane foam as a thermal building insulator at varied mixing conditions. In *IOP Conference Series: Earth and Environmental Science,* **2023**, 1201(1), 12057.

[47] Shivakumar, N.D., Deb, A. Dependence of the mechanical properties of rigid PU foam on density. *Journal of Reinforced Plastics and Composites*, **2022**, 41(9-10), 355-363.

[48] Nisa, R.S., Munawar, S.S., Hermawan, D., Purnomo, D., Subiyanto, B., Ismadi, A.S., Poniman, D. To the Physical and Mechanical Properties of Rigid Polyurethane Foam Composite. In *Proceedings of the 1st International Conference of Lignocellulose: ICON-LIG 2021*, **2025**, 175.

[49] Chaudhary, S., Chaudhary, M.L., Gupta, R.K. Bio-Derived Chemicals for Polyurethane Foams. In Bio-Based Polymers: Farm to Industry. *American Chemical Society*, **2024**, 1, 121-141.

[50] Jayalath, P. Assessing the Environmental Life Cycle Impacts of Lignin-Derived Polyurethane Foams for High-Performance Insulation: A Life Cycle Assessment Approach. *Master's thesis, State University of New York College of Environmental Science and Forestry*, **2024**, 23-41.

[51] Jabar, J., Fakinlede, F., Bello, B. Synthesis of rigid polyurethane foam using polyols from fractionated locally sourced vegetable oil. *FUW Trends In Science & Technology Journal*, **2018**, 3(2), 384-388.

[52] He, P., Ruan, H., Wang, C., Lu, H. Mechanical properties and thermal conductivity of thermal insulation board containing recycled thermosetting polyurethane and thermoplastic. *Polymers*, **2021**, 13(24), 4411.

[53] Deshmukh, M., Yadav, M. Optimizing thermal efficiency of building envelopes with sustainable composite materials. *Buildings*, **2025**, 15(2), 230.

[54] Carriço, C.S., Fraga, T., Pasa, V.M. Production and characterization of polyurethane foams from a simple mixture of castor oil, crude glycerol and untreated lignin as bio-based polyols. *European Polymer Journal*, **2016**, 85, 53-61.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.